

Northwind Robotics

Orion OR-500 Autonomous Warehouse Robot — Operator Manual

Document revision 2.1, published June 2025. This manual covers the Orion OR-500 heavy-payload model only. For the compact Atlas AR-200 platform, refer to the separate Atlas operator manual.

1. Overview

The Orion OR-500 is a heavy-payload autonomous mobile robot designed for pallet movement in warehouses and distribution centres. It was first released in 2024. Like the Atlas AR-200, the Orion navigates using LiDAR and floor-mounted QR fiducials, but it is substantially larger and stronger.

Each Orion OR-500 carries a maximum payload of 500 kilograms and travels at a top speed of 1.2 metres per second when carrying a load. Without a load it may travel at up to 1.6 metres per second. Unlike the Atlas, the Orion has a minimum turning radius of 0.8 metres due to its fixed rear axle.

2. Technical Specifications

The OR-500 is powered by a 48-volt lithium-iron-phosphate battery pack with a capacity of 60 ampere-hours. A full charge takes approximately 3 hours and provides up to 8 hours of continuous operation under typical load. The robot weighs 340 kilograms unladen. Its onboard compute runs on an ARM-based controller with 16 gigabytes of memory.

The operating temperature range is 0 to 45 degrees Celsius. The OR-500 is rated IP65 and, unlike the Atlas AR-200, is suitable for cold-storage environments down to 0 degrees. Every unit carries a serial number of the form OR5-XXXXXX printed on the rear chassis plate.

3. Charging and Battery Care

The Orion OR-500 returns to its charging dock automatically when its battery level falls below 20 percent. Because of its larger battery, the OR-500 uses a high-current dock that is not compatible with the Atlas AR-200 charging dock. The two robot models therefore require separate docking infrastructure.

The battery pack is rated for 3,000 full charge cycles before its capacity drops below 80 percent. Replacement battery packs are sold under part number NW-BAT-48-60. Note that this is a different part from the Atlas battery; the two are not interchangeable.

For long-term storage, charge the battery to approximately 50 percent and store the robot between 10 and 25 degrees Celsius. Because of its weight, the OR-500 must be stored on a level surface rated for at least 500 kilograms.

4. Safety

The OR-500 is fitted with 360-degree safety scanners. Because of its greater mass and stopping distance, its protective field extends 2.0 metres ahead at full speed, compared with 1.2 metres for the Atlas AR-200. The emergency stop button is located on both the front and rear surfaces of the robot.

5. Error Codes

The Orion OR-500 shares the Northwind common error-code scheme with the Atlas AR-200, but adds several codes specific to its heavier drivetrain and lifting mechanism. Codes below E500 have the same meaning across both models.

Code	Meaning	Operator action
E101	Battery temperature too high	Move robot to cooler area; wait 30 minutes
E203	Safety scanner obstructed	Clear the scanner window
E415	Payload exceeds limit	Remove excess load below 500 kg
E610	Lift mechanism jammed	Lower load manually; contact support
E620	Pallet not detected	Reposition robot square to the pallet
E630	Load unstable during transit	Stop; re-centre the load on the deck

As with the Atlas, unlisted error codes should be reported to robotics-support@northwind-example.com with the robot serial number.

6. Routine Maintenance

Inspect the Orion OR-500 at the start of each shift, paying particular attention to the lifting deck and its hydraulic seals. Every 500 operating hours, inspect the drive wheels; because of the heavier payload, Orion drive wheels wear faster than Atlas wheels and are sold under part number NW-WHL-500.

The lifting mechanism must be serviced every 2,000 operating hours. A full preventive-maintenance service is recommended every 3,000 operating hours or once per year, whichever comes first, and must be carried out by a Northwind-certified technician to keep the warranty valid.

7. Firmware

The Orion OR-500 runs the same fleet firmware family as the Atlas but on a different build. As of this revision, the current stable firmware version for the OR-500 is 2.9.3. Note that Atlas firmware version 4.7.1 must not be installed on an Orion; the fleet console prevents this automatically.

8. Fleet Integration

The Orion OR-500 is managed by the same Northwind fleet management console as the Atlas AR-200, and the two models can operate in the same warehouse under a single console. However, because the Orion is heavier and slower, the console assigns it to wider aisles by default. Aisles used by the OR-500 must be at least 2.4 metres wide.

A single console can manage up to 100 robots of mixed models. If a mixed fleet exceeds 60 Orion units, Northwind recommends a second wireless controller for load balancing.

9. Warranty Summary

The Orion OR-500 is covered by Northwind's standard hardware warranty. In brief, the OR-500 carries a three-year warranty on the robot chassis and a two-year warranty on the battery pack — longer than the Atlas AR-200 because of the Orion's industrial duty rating. Full terms are set out in the Northwind Robotics Warranty and Support Policy, which takes precedence over this summary.